Notes: Bergeaud, Schmidt, and Zago (JFE, 2026)

Patents that match your standards: Firm-level evidence on competition, innovation and growth

Contents

1 Big picture

- **Question.** When an industry chooses a new technical standard, which firms benefit, how does the standard change competition and innovation in the short run, and does it ultimately raise long-run growth?
- **Setting.** The paper combines a huge patent-level data set with a huge standards data set. It matches the semantic content of patents to the content of standard documents and then aggregates those patent-to-standard links to the firm-quarter level.
- **Core challenge.** A firm may look "close" to a standard either because it genuinely developed the technology that later became codified, or because it simply has a large patent portfolio, lobbied in the standard-setting process, or operates in a technology field that mechanically overlaps with the standard. The paper has to separate these stories.
- **Main measurement contribution.** The paper builds a new firm-level measure of **technological proximity to standards** by matching 7.5 million US patents to about 0.65 million standards using TF-IDF weighted cosine similarity.
- **Main empirical result at the firm level.** Firms whose patent portfolios are closer to a newly released standard earn abnormal returns when the standard's content becomes public, see upward EPS forecast revisions, and then experience temporary increases in sales and market share.
- **Competition and innovation result.** After standardization, firms close to the new standard increase R&D only when they operate in competitive industries. In non-competitive industries, the R&D response is weak or negative. This is consistent with an inverted-U style link between competition and innovation.
- **Aggregate result.** The leaders gain first, but the effect does not last. Over time, followers catch up through spillovers, increase R&D and patenting, and drive higher sectoral growth and lower concentration.
- **Why this paper matters.** It connects the industrial-organization literature on standard setting to macro questions about competition, innovation, diffusion, and growth. The paper's main message is that standards are not just about rewarding current leaders; they also create a common technological basis that helps laggards catch up.

2 Data and measurement (key objects)

2.1 Patent and standards universe

- **Patent data.** The paper uses all US priority patent applications in IFI CLAIMS from 1980 to 2020, with the estimation sample restricted to patents filed between 1980 and 2010. This yields more than 7.5 million patent-level observations.
- **Standards data.** Standards come from the Searle Center database, built largely from Perinorm. After collapsing equivalent documents across organizations and countries, the authors obtain about 0.65 million standards.

- **Patent–standard matches.** The semantic matching procedure generates more than 1.1 billion patent–standard pairs; for computational reasons the authors retain the best 100 million pairs and then keep matched and unmatched patents filed before the standard release.
- **Key descriptive facts.** In the final patent-standard sample, the mean similarity score is 0.24, the mean time lag from patent filing to subsequent standard release is 12.36 years, and the average patent matches 7.65 standards from its filing date onward. Importantly, about 75% of patents do not match any standard at all.

2.2 Firm-level panel

- **Firm mapping.** Patents are linked to publicly listed firms using the Kogan et al. crosswalk from patent assignee to Compustat firm.
- **Balance-sheet variables.** The main firm-level outcomes are sales, R&D, and market share. Sales and R&D are normalized by the firm's mean fixed assets. Market share is defined within NAICS 3-digit industries.
- **Controls.** The paper controls for age, Tobin's q, leverage, and market capitalization. It also uses firm-level markups and industry markup dispersion to classify markets as more or less competitive.
- **Financial-market variables.** Abnormal returns are computed from a CAPM-style regression that uses a NAICS3 industry portfolio as the market portfolio. The paper also matches IBES data to measure one-year-ahead EPS forecasts.
- **Sample.** The final firm-quarter sample contains 24,162 observations from 892 publicly listed, nonfinancial, nonutility patenting firms between 1984 and 2010.
- **Summary statistics.** The mean firm-quarter proximity measure is 0.46 with standard deviation 2.24; 48% of observations have positive proximity. The average firm has sales equal to 0.63 of mean fixed assets, R&D equal to 0.05, market share of 0.05, age of about 99 quarters, q of 1.98, leverage of 0.19, and market capitalization of about $9.18 billion.

2.3 Timing, treatment, and economic meaning

- **Standards timing.** The authors only observe the official publication date of a standard, but knowledge of the approval procedure lets them back out an *imputed* public-release window in which the draft becomes publicly known. This is roughly 4 to 8 months before final publication, that is, around quarter -2 relative to the official date.
- **Why this matters.** The imputed public-release window is central to identification: if the content of the standard is genuinely new information, financial markets and firms should react around that window, not long before.
- **Treatment variable.** The firm-level treatment is $\text{Tech.Prox}_{i,t}$, a continuous measure of how close firm i's stock of patents is to the standards released in quarter t.
- **Repeated treatment.** A firm can be close to one standard in one quarter, not close in a later quarter, and then close again to another standard later on. So this is not a once-and-for-all treatment setting; it is a repeated-treatment, varying-intensity setting.

3 Models and empirical specifications (all equations + interpretation)

3.1 Semantic matching of patents to standards

The paper constructs a bilateral similarity score between standard i and patent j using TF-IDF weighted cosine similarity:

$$\text{Score}(i,j) = \frac{\mathcal{A}(i) \cdot \mathcal{B}(j)}{\|\mathcal{A}(i)\| \, \|\mathcal{B}(j)\|} = \frac{\sum_{k=1}^{K} \mathcal{A}(i)_k \mathcal{B}(j)_k}{\sqrt{\sum_{k=1}^{K} \mathcal{A}(i)_k^2} \sqrt{\sum_{k=1}^{K} \mathcal{B}(j)_k^2}}.$$

The vectors $\mathcal{A}(i)$ and $\mathcal{B}(j)$ summarize the weighted occurrence of cleaned and stemmed k-grams in the standard and in the patent abstract.

- **Interpretation.** High scores mean that the patent's semantic content overlaps strongly with the content of the standard.
- **Why TF-IDF matters.** Rare but informative terms get more weight, so the score focuses on technologically meaningful overlap rather than generic language.
- **Practical restriction.** The authors keep only keywords that do not appear too often across documents, which sharpens the match and reduces noise.

3.2 Firm-level technological proximity to a standard

The core firm-level variable is an IPC-weighted, depreciated aggregation of patent-standard scores:

$$\text{Tech.Prox}_{i,t} = \sum_{j \in J} \omega_{i,j,t_0} \sum_{p \in j} \zeta^{(t-t_p)} \, \text{Score}_{i,p,j,t},$$

where ω_{i,j,t_0} is firm i's presample patent share in IPC class j, t_p is the filing date of patent p, and $\zeta = (1-\delta)$ is a depreciation factor with annual depreciation rate set to 20%.

- **Interpretation.** This is a firm-quarter measure of how close the firm's *existing stock of knowledge* is to the technologies codified in newly released standards.
- **Why presample IPC weights are used.** They reduce the importance of patent classes outside the firm's core technological domain and help limit endogenous self-selection into fields that later become standardized.
- **Why depreciation is used.** It downweights old patents, helps prevent the measure from being driven mechanically by portfolio size, and makes it less likely that the paper is merely capturing technologies that were already fully diffused before standardization.

3.3 Patent-level validation regressions

To show that the semantic score captures an economically meaningful dimension of patent quality, the paper relates patent scores to patent value, forward citations, and renewal decisions. For patent value, the main regression is

$$\log(\text{value}_i) = c + \alpha \, \text{score}_i + \beta \log(1 + \text{cit}_i) + f_{t(i),k(i)} + \varepsilon_i,$$

where value_i is the Kogan et al. market-value measure for patent i, cit_i is forward citations, and $f_{t(i),k(i)}$ are grant-time by IPC fixed effects.

- **Interpretation.** The score is not just a text object; it is positively related to the economic, scientific, and private value of patents.
- **Empirical takeaway.** Patents with higher scores are more valuable, more cited, and less likely to be allowed to expire.
- **Important conceptual point.** The score is not simply duplicating Kogan et al. value. It measures something slightly different: the patent's ability to fit, deploy, and diffuse under a newly codified industry standard.

3.4 Distributed lead–lag design for firm-level outcomes

The main empirical design is a distributed lead–lag model:

$$Y_{i,t} = \alpha_i + \phi_{s(i),t} + \sum_{n=-16}^{12} \beta_n \,\text{Tech.Prox}_{i,t+n} + X'_{i,t-1}\eta + \varepsilon_{i,t}.$$

Here $Y_{i,t}$ is the firm-level outcome, α_i is a firm fixed effect, $\phi_{s(i),t}$ is a NAICS3-industry-by-time fixed effect, and $X_{i,t-1}$ contains lagged controls.

- **Interpretation.** The coefficient β_n traces how being close to the standard released at $t+n$ affects the firm's outcome at time t.
- **Why leads are included.** They test for pretrends and anticipation. If firms or markets already knew the content of the standard far in advance, future-treatment coefficients would move before the release.
- **Why lags are included.** They recover the dynamic response after standardization and avoid bias from overlapping standard releases.
- **Why this design is appropriate.** Treatment is repeated and continuous. The model allows treatment intensity to switch on and off over time and therefore fits this setting better than a one-shot treatment design.

3.5 Abnormal returns and market expectations

To construct abnormal returns, the paper estimates a rolling 10-year industry-level CAPM for each firm:

$$r_{i,t} - r_{f,t} = \alpha_{i,\tau} + \beta_{i,\tau}(r_{s,t} - r_{f,t}) + \varepsilon_{i,t}, \qquad t \in (\tau - 40, \tau],$$

and defines abnormal return at $\tau + 1$ as

$$ar_{si,\tau+1} = (r_{i,\tau+1} - r_{f,\tau+1}) - \left(\hat{\alpha}_{i,\tau} + \hat{\beta}_{i,\tau}(r_{s,\tau+1} - r_{f,\tau+1})\right).$$

It also studies revisions in one-year-ahead EPS forecasts from IBES:

$$\Delta E[EPS_{i,t+4}] = E[EPS_{i,t+4} \mid \mathcal{I}_t] - E[EPS_{i,t+4} \mid \mathcal{I}_{t-1}].$$

- **Interpretation.** These two variables ask whether standardization contains new information for markets.
- **Economic logic.** If standardization merely rubber-stamps technologies that everyone already understands, neither abnormal returns nor forecast revisions should move when the draft becomes public.

3.6 Sales, market share, and R&D

The same distributed lead–lag design is used with three firm-level outcomes:

- normalized sales,
- firm-level market share in NAICS3,
- the four-quarter moving average of R&D expenditure.

The market-share regression distinguishes a pure demand-expansion story from a competition story. The R&D regressions are then split by ex ante market structure, using markup dispersion to classify industries as competitive or non-competitive.

- **Interpretation.** Standards do not just increase firm sales because the whole market expands. They temporarily shift market shares toward firms whose technologies fit the new standard.
- **Competition mechanism.** In competitive markets, the temporary lead induced by standardization is worth defending, so frontier firms invest more in R&D. In non-competitive markets, this incentive is weaker.

3.7 Aggregate growth decomposition

At the sector level, the paper classifies firms with $\text{Tech.Prox}_{i,t} > 0$ as **leaders** and the rest as **followers**. It then aggregates sales growth to the NAICS3 industry-quarter level and asks how much sectoral growth is explained by each group.

- **Interpretation.** This turns the firm-level mechanism into an industry-level decomposition: leaders gain first, but followers may matter more later.
- **Key question.** Does standardization generate persistent leader dominance, or does it create spillovers that let followers catch up and eventually drive aggregate growth?

3.8 Convergence to the standard

To study post-standard diffusion, the paper estimates at the patent level

$$\log\left(\frac{\text{score}_{i,j,s}}{\text{score}^{Best}_{t^*<\tau,s}}\right) = \alpha + \beta \log\bigl(\text{score}^{Best}_{t^*<\tau,s}\bigr) + \gamma\, time_{i,s} + \delta_{t(j)} + \varepsilon_{i,j,t},$$

for patents issued before and after the publication of standard s at time τ.

- **Objects.** $\text{score}_{i,j,s}$ is the score of patent i with standard s; $\text{score}^{Best}_{t^*<\tau,s}$ is the best pre-publication score in the preceding five years.

- **Interpretation.** If newer patents converge toward the new standard, the relative distance to the best pre-standard patent should shrink after publication.
- **Frontier-firm heterogeneity.** The paper further interacts this regression with an indicator for firms that already had at least one positive-score patent before the standard was released, to distinguish early leaders from laggards.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the paper is not just measuring ex post standardization

- The first identification problem is whether the paper is simply capturing technologies that were already widely diffused before the standard was published.
- The paper's answer is the financial-market event study: abnormal returns and EPS forecast revisions move only when the standard's content becomes public, not earlier.
- That timing result is central. It suggests the content of the standard is genuinely informative and that the standard release is a real coordination event rather than a purely ceremonial acknowledgment of an already dominant technology.

4.2 What identifies the firm-level effects

- **Within-firm variation.** Identification comes from changes over time in a firm's exposure to newly released standards.
- **Industry-time fixed effects.** The NAICS3-by-time fixed effects absorb any common demand, legislation, seasonality, momentum, or macro shocks that hit all firms in the same industry-quarter.
- **Lead–lag structure.** Including leads and lags isolates the full dynamic response and avoids confounding from previous or subsequent standard releases.
- **Public-release window.** Because the draft becomes public before the official publication date, the event-study design can distinguish the information release from later real adjustments.

4.3 Validation against lobbying, strategic patenting, and measurement error

The paper spends substantial effort showing that $\text{Tech.Prox}_{i,t}$ captures the *quality and relevance* of a firm's patent portfolio rather than mechanical correlates.

- **SSO participation.** Adding controls for SSO membership around the event does not eliminate the results, and the membership coefficients themselves are not significant.
- **International versus US SSOs.** Recomputing the proximity measure using only international standards leaves the results intact, while using only US-SSO standards weakens them. This is consistent with the concern that lobbying is more plausible in national SSOs and strengthens the paper's preferred interpretation.

- **Strategic patenting.** Controlling for the number of patents filed in the year before standardization does not alter the results, so the findings are not driven by a pre-event patenting surge.
- **Top innovators.** Excluding the top 25% most innovative firms still leaves the baseline patterns intact.
- **Portfolio-size decomposition.** The extensive margin of patenting can explain some raw variation in log(Tech.Prox), but once firm and industry-time fixed effects are added, portfolio size explains only about 5% of the variation; the remainder comes from intensive or quality margins.
- **Top-match redefinition.** Restricting the measure to only the highest-quality patent–standard matches preserves the results.
- **Placebo randomization.** Randomly reassigning standard scores across patents and rebuilding Tech.Prox produces flat, insignificant event-study coefficients. So merely having many patents is not enough.
- **Portfolio similarity clusters.** Comparing firms within technologically similar patent-cluster groups still delivers the main results.
- **Ex ante characteristics.** Conditional on firm fixed effects, firms with positive proximity do not look systematically different four quarters before release in measures such as size, age, leverage, ROE, cost of capital, and valuation.

4.4 Relation to the literature

- **Standardization.** The paper takes a classic industrial-organization topic—technology standards and SSOs—and brings it into a firm-level and macro-growth framework.
- **Competition and innovation.** The heterogeneous R&D response across competitive and non-competitive industries is tightly connected to the Aghion et al. inverted-U logic.
- **Spillovers and diffusion.** The long-run follower catch-up results connect the paper to the literature on spillovers, diffusion, and frontier convergence.
- **Text analysis in economics.** Methodologically, the paper contributes a large-scale semantic matching procedure between patents and standards, rather than between patents and citations only.

5 Tables and figures

5.1 Figure 1: timing of standards approval

Read:

- The figure explains the paper's event time. Official publication is quarter 0, but the *draft* becomes public about two quarters earlier.
- This is why the red shaded area in the event studies matters: it is the window in which the standard's content first becomes observable.

- The figure is essential for interpreting the market reaction. Markets should move when the content becomes public, not necessarily at the official publication date.

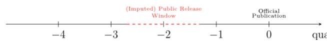


Fig. 1. The Timing of standards approval.
Notes: This figure sketches the administrative procedure of standards' approval. The official date of publication of the standard by the standard-setting organization is known and occurs at quarter 0. Given information on the administrative procedure of approval and publication of a standard, we back up the (imputed) time window in which the judging committee voted in favor of the standard and made the standard's draft publicly available. This happens roughly around −2, i.e. 2 quarters before the official publication date. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Descriptive statistics of the matching procedure.

	Mean	*SD*	*Min*	*Max*	*p1*	*p5*	*p25*	*p50*	*p75*	*p95*	*p99*	*N*
Final sample:	unmatched and matched patents with filing year < standard release year											
Score	0.24	0.15	0.00	1.00	0.00	0.00	0.16	0.21	0.29	0.50	0.73	64,488,773
Time lag	12.36	8.69	0.00	38.00	0.00	1.00	5.00	11.00	18.00	29.00	34.00	57,556,532
Standards	7.65	47.48	0.00	1778.00	0.00	0.00	0.00	0.00	0.00	22.00	224.00	7,521,262

Notes: The table reports descriptive statistics for the score, the number of matched standards per patent and the time lag (in years) between the release of the standard and the filing of the patent. The sample only comprises utility patents, matched and unmatched, for which the patent filing date does not exceed the standard release date. The number of observations in the row reporting time lag statistics is lower than in the row reporting score statistics, as the latter also includes unmatched patents for which, by definition, no time lag data is available.

5.2 Table 1: descriptive statistics of the matching procedure

Read:

- The average patent–standard similarity score is 0.24, but the distribution is highly skewed; most patents do not match standards well at all.
- The mean time lag between patent filing and later standard release is 12.36 years, which is consistent with the idea that standardization often codifies technology long after invention.
- The mean number of matched standards per patent is 7.65, but the median is zero. This underscores that meaningful matches are rare and concentrated.
- This table helps explain why the authors worry that any useful firm-level measure must be driven by a small set of high-quality matches rather than by blanket overlap.

5.3 Table 2: descriptive statistics for the firm-level sample

Read:

- The average firm-quarter proximity measure is 0.46, but the median is zero, and 48% of observations have positive proximity.
- The average firm has sales equal to 63% of mean fixed assets and R&D equal to 5%.
- The average firm-level market share is about 5%, average market capitalization is about $9.18 billion, and average markup is 1.46.
- Roughly one-quarter of firms operate in industries classified as non-competitive using markup dispersion.

Table 2
Descriptive statistics for the firm-level data.

	Mean	*SD*	*p1*	*p5*	*p25*	*p50*	*p75*	*p95*	*p99*	*N*
(A) Proximity Measure										
Tech.Prox (inpercent)	0.46	2.24	0.00	0.00	0.00	0.00	0.18	1.94	7.87	24,162
$\mathbb{I}[Tech.Prox > 0]$	0.48	0.50	0.00	0.00	0.00	0.00	1.00	1.00	1.00	24,162
(B) Firm Characteristics										
Sales	0.63	0.72	0.01	0.07	0.25	0.47	0.78	1.60	2.99	24,162
R&D	0.05	0.29	0.00	0.00	0.00	0.01	0.05	0.18	0.59	24,162
Market Share (NAICS3)	0.05	0.10	0.00	0.00	0.00	0.01	0.05	0.21	0.49	24,162
Age (quarters)	98.99	49.92	21.00	21.00	53.00	110.00	137.00	171.00	181.00	24,162
Q	1.98	2.15	0.74	0.90	1.17	1.50	2.12	4.44	8.69	24,162
Leverage	0.19	0.16	0.00	0.00	0.06	0.17	0.27	0.45	0.65	24,162
Market Cap. (Billion$)	9.18	29.00	0.00	0.02	0.19	1.27	5.61	42.21	139.89	24,162
Firm Markup	1.46	0.69	0.88	0.98	1.09	1.25	1.57	2.69	4.55	24,162
Industry Markup Dispersion	0.55	0.33	0.14	0.16	0.27	0.48	0.79	0.99	1.44	24,162
$\mathbb{I}(Non-CompetitiveIndustry)$	0.26	0.44	0.00	0.00	0.00	0.00	1.00	1.00	1.00	24,162
(C) Financial Mkts										
ar^{NAICS3}	0.00	0.40	−0.58	−0.33	−0.09	0.00	0.09	0.28	0.57	18,531
1*yr EPS Forecast* ($)	1.68	1.72	−1.07	0.11	0.67	1.32	2.23	4.68	8.41	17,032

Notes: The variable *Tech.Prox* measures the proximity of the portfolio of patent of the firm to the standard. $\mathbb{I}[Tech.Prox > 0]$ is a dummy that takes value one for positive values of the variable *Tech.Prox*. *Sales* is the firm-level of sales normalized by the mean-level of fixed assets (property, plant, equipment). The *Market Share* is constructed at the NAICS 3-digit level. *R&D* is the level of R&D expenditure normalized by the mean-level of fixed assets. *Age* is the number of quarters the firm is active. *Q* is the q-value of investments, and is built as the value of liabilities plus the market value of common equity divided by the book value of assets. *Leverage* is debt over the book value of assets. *Market Capitalization* is expressed in Billion of US dollars. The NAICS 3-digit industry markup is constructed following De Loecker et al. (2020). $\mathbb{I}(Non-CompetitiveIndustry)$ is a dummy that takes value one if a firm is operating in a NAICS 3-digit industry with markup above the 75th percentile. ar^{NAICS3} is a measure of stock market abnormal return built from a standard CAPM model with a NAICS 3-digit index as market portfolio. The *1yr EPS Forecast* is the mean forecast across all professional forecasters of the earning-per-share expected by the end of the following fiscal year, and it is expressed in dollars.

Table 3
Regression results for financial, scientific and private patent value.

	(1)	(2)	(3)	(4)	(5)
	Kogan et al. (2017)		*Forward citations*	*Expiration*	
Score	10.8471***	10.6565***	3.4686***	−1.5103***	−1.2043***
	[0.299]	[0.289]	[0.082]	[0.074]	[0.059]
Observations	2,283,666	2,283,666	4,374,892	8,806,908	8,806,908

Notes: Patent-level regressions. "Score" is the sum of scores across all associated standards of a given patent (see Section 3.1), normalized to take values between 0 and 1. The dependent variables are: columns 1 and 2: the economic value of the patent as computed by Kogan et al. (2017); columns 3: the number of forward citations received by the patent over a 10 year window; columns 4-5: maintenance decisions for patent *i*. Columns 1-2 use an OLS estimator, column 3 uses a Poisson model and columns 4 and 5 use a Cox hazard model. Columns 1-2 include a technological class (3-digit IPC) interacted with the granting year/quarter fixed effect as in Kogan et al. (2017). Column 3 includes the 3-digit IPC class interacted with the filing year/quarter fixed effect. Similarly, the Cox model in columns 4 and 5 stratifies the data by the interaction of the 3-digit IPC class with the filing year/quarter. Columns 2 and 5 additionally control for the logarithm of the number of forward citations received by the patent. The sample of patents includes: columns 1 and 2: priority patents from IFI claims matched to the Kogan et al. (2017) sample with filing year between 1980 and 2010; column 3: priority patents from IFI claims published over the period 1980–2010; columns 4 and 5: priority patents from IFI claims matched to USPTO patents with information on renewal fees for the period 1981–2015. In the last two columns, a patent can be included several times in the sample due to different schedules of renewal (see Section 3.3). "*", "**" and "***" designate significance at the 1%, 5% and 10% level.

5.4 Table 3: patent value, citations, and patent renewal

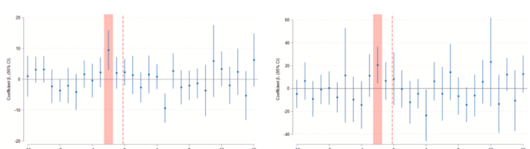
Read:

- The score strongly predicts Kogan et al. patent value: the coefficient is about 10.85 without a citation control and 10.66 with the control.
- The score also predicts scientific value: the forward-citation coefficient is 3.47 in the Poisson specification.
- The score predicts private value too: the Cox hazard coefficients are negative, around −1.51 and −1.20, meaning patents associated with later standards are less likely to expire.
- Quantitatively, the authors estimate that matched patents are worth about 8.6% more; with mean score 0.008, this corresponds to roughly \$619,000 at the median and \$2.1 million at the mean.

5.5 Figure 2: abnormal returns and EPS revisions

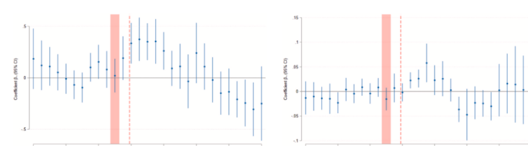
Read:

- There is no common pretrend before the draft becomes public.
- At about quarter −2, firms closer to the standard earn positive abnormal returns and experience positive revisions in one-year-ahead EPS forecasts.
- This is the paper's cleanest evidence that markets learn something new from the standard release itself.
- The figure therefore validates the core interpretation that standards convey new information beyond what was already known from firms' patent portfolios.



(a) ar^{NAICS3} (b) $\Delta\mathbb{E}(EPS)$

Fig. 2. Technological proximity and financial markets' reaction.
Notes: Fig. 2(a) plots the estimated coefficients of Eq. (2) (see Section 4.1) when the dependent variable is the firm-level abnormal return computed through the CAPM model with market portfolio defined at the NAICS3 industry-level. Fig. 2(b) plots the estimated coefficients when the dependent variable is the change in the 1-year EPS forecast. See Section 2.3 for more information on variables construction. In both figures, the 95% confidence intervals for each point-estimate is reported. Standard errors are clustered at the firm level. The red area indicates the imputed time window of public release of the standard's content, based on knowledge of the procedure of approval. The red-dashed line indicates the official publication of the standard, as reported by the standard-setting organization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) Sales (b) Market Share

Fig. 3. Technological proximity, sales and market share.
Notes: Fig. 3(a) and 3(b) plot the estimated coefficients of Eq. (2) (see Section 4.1) when the dependent variable is respectively the level of sales (normalized by the mean-level of fixed assets) and the firm-level market share defined at NAICS3 industry-level. See Section 2.3 for more information on variables construction. In both figures, the 95% confidence intervals for each point-estimate is reported. Standard errors are clustered at the firm level. The red area indicates the imputed time window of public release of the standard's content, based on knowledge of the procedure of approval. The red-dashed line indicates the official publication of the standard, as reported by the standard-setting organization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.6 Figure 3: technological proximity, sales, and market share

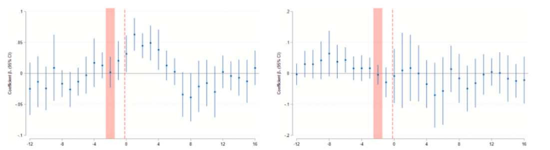
Read:

- Firms closer to the new standard sell more after publication, with the effect lasting about five quarters.
- The same dynamic appears for market share, so the effect is not just overall market expansion. It is also a reallocation of market share toward firms that can implement the standard quickly.
- Using the extensive-margin indicator $I[\text{Tech.Prox} > 0]$, the authors estimate a maximum temporary increase of about 2.0% in both sales and market share for firms close to the standard.

5.7 Figure 4: technological proximity and R&D

Read:

- In competitive industries, firms close to the standard increase R&D after the standard is published, and the effect lasts for about five quarters.
- In non-competitive industries, the R&D response is weak and points negative.
- For the full sample, the dominant effect is still an increase in R&D. Using the extensive-margin treatment, the largest estimated increase is about 4.8%.
- This is the clearest evidence that the standardization event temporarily changes the intensity of competition.



(a) R&D (Competitive Ind) (b) R&D (Non-Competitive Ind)

Fig. 4. Technological proximity and R&D.
Notes: Fig. 4(a) and 4(b) plot the estimated coefficients of Eq. (2) (see Section 4.1) when the dependent variable is the 4-quarter moving average of R&D expenditure (normalized by the mean-level of fixed assets) and the sample is composed respectively by firms operating in a competitive and non-competitive industry. See Section 2.3 for more information on variables construction. In all figures, the 95% confidence intervals for each point-estimate is reported. Standard errors are clustered at the firm level. The red area indicates the imputed time window of public release of the standard's content, based on knowledge of the procedure of approval. The red-dashed line indicates the official publication of the standard, as reported by the standard-setting organization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
Aggregate effects on growth.

	Industry	*Leaders*	*Followers*
Mean Sectoral Growth Rate (%)	1.64	1.05	0.58
(1yr – Cumulative) Change in Growth due to Mean Sectoral Shock (pp)	−0.05 (0.04)	0.05 (0.03)	−0.10 (0.06)
(4yr – Cumulative) Change in Growth due to Mean Sectoral Shock (pp)	0.19 (0.10)	0.02 (0.04)	0.18 (0.07)

Notes: The first line of this table shows the average sectoral growth and its decomposition between leaders and followers. The second and third line show the cumulative effect of the introduction of a standard on sectoral growth respectively one and four year after the official publication of the standard. Standard errors are in parenthesis. See Appendix F for details on data and estimation.

5.8 Table 4: aggregate effects on growth

Read:

- The average industry grows by 1.64% per quarter. Leaders account for 1.05 percentage points of that, followers 0.58.
- Over the first year after the standard's introduction, average sectoral growth falls by 0.05 percentage points: leaders gain 0.05 points, but followers lose 0.10. This is the short-run reallocation effect.
- Over four years, the sign reverses. Sectoral growth rises by 0.19 percentage points, driven almost entirely by followers (+0.18), while leaders contribute little (+0.02).
- This table summarizes the paper's main aggregate message: leaders gain first, followers drive long-run growth.

5.9 Figure 5: aggregate effects of standardization

Read:

- Followers' market share initially falls, but from about the third year onward rises persistently.
- Followers' share of industry R&D and followers' share of new patents both rise in the long run, showing that catch-up happens through innovation rather than only through reallocation of existing output.

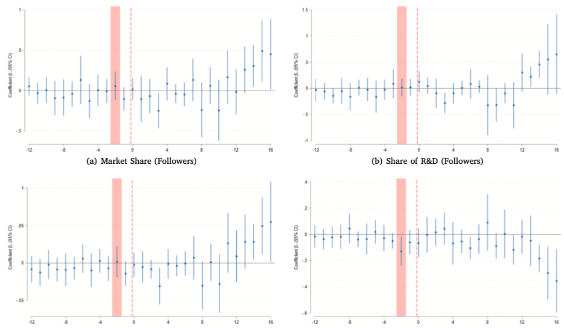


Fig. 5. Aggregate effects of standardization.
Notes: Figs. 5(a)–5(b) plot the estimated coefficients when the dependent variable is respectively: the market share of followers in the NAICS3 industry, the followers' share of total R&D expenditure in the NAICS3 industry, the followers' share of total patents issued in the NAICS3 industry, the sectoral (normalized) Herfindahl–Hirschman Index (HHI). See Appendix F for more details on data construction and estimation. In all figures, the 95% confidence intervals for each point-estimate is reported. Standard errors are double-clustered at (NAICS3) industry-level and date. The red area indicates the imputed time window of public release of the standard's content, based on knowledge of the procedure of approval. The red-dashed line indicates the official publication of the standard, as reported by the standard-setting organization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- The Herfindahl–Hirschman Index falls in the long run, meaning concentration declines as followers catch up.
- The figure shows that standardization ultimately promotes diffusion and competition rather than permanent monopoly.

5.10 Table 5 and Figure 6: convergence toward the standard and citations from followers

Table 5
Convergence towards the new standard.

	(1)	(2)	(3)	(4)	(5)	(6)
	Pre-standard Release		*Post-standard Release*			
$log(score^{Best}_{t^*<\tau,s})$	−0.0018	0.0136	−0.0895***	−0.0892***	−0.1897***	−0.2209***
	(0.0091)	(0.0093)	(0.0096)	(0.0096)	(0.0080)	(0.0089)
$log(score^{Best}_{t^*<\tau,s}) \times \mathbb{I}(Frontier\ Firm)$						0.1501***
						(0.0130)
Observations	334,608	334,608	383,341	383,341	383,341	383,341
Patent filing date Fe	Yes	Yes	Yes	Yes	Yes	Yes
N. of qtr to/from Standard Pub.	No	Yes	No	Yes	Yes	Yes
Frontier Firm	–	–	No	No	Yes	Yes

Notes: Patent-level regressions. The dependent variable is the logarithm of the ratio between the variable $score_{i,s}$ and $score^{Best}_s$ which respectively represent the score obtained from matching i with standard s, $score^{Best}_s$ whereas is the highest score realized by the best patent in the 5 years prior to the publication of standard s. $time$ is the number of quarters intercurrent before or after the publication of the standard. $\mathbb{I}(Frontier\ Firm_i)$ indicates whether patent i belongs to a firm that was already close to the standard prior to its publication. "*", "**" and "***" designate significance at the 1%, 5% and 10% level.

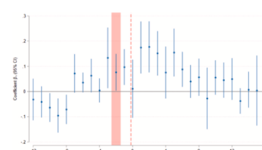


Fig. 6. Patent citations from followers.
Notes: Fig. 6 plots the estimated coefficients of Eq. (2) when the dependent variable is the share of the number of patent citations leaders receive from laggard firms. The 95% confidence intervals for each point-estimate is reported. Standard errors are clustered at the firm level. The red area indicates the imputed time window of public release of the standard's content, based on knowledge of the procedure of approval. The red-dashed line indicates the official publication of the standard, as reported by the standard-setting organization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Read:

- Before publication, there is no convergence: the coefficient on the best pre-standard score is statistically insignificant.
- After publication, the coefficient becomes strongly negative (about −0.089), meaning newer patents begin to converge toward the new standard only once the standard is public.
- The effect is stronger when the patent comes from firms that were not already frontier firms. The estimates imply convergence of roughly 22% for non-frontier firms versus about 7% for frontier firms.
- Figure 6 adds direct spillover evidence: firms whose patents match the standard well are increasingly cited by laggard firms.

6 Short summary

- **Measurement.** The paper constructs a new semantic measure of firm proximity to newly released technology standards by matching patent abstracts to standard documents.
- **Short-run economics.** Firms closer to the new standard receive good news in financial markets and then temporarily gain sales and market share. They also raise R&D if their market is competitive.
- **Long-run economics.** Standards help laggards catch up through spillovers, patenting, and convergence to the new technological frontier. As a result, sectoral growth rises and concentration falls in the long run.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that standardization is an economically meaningful coordination event.
- It rewards firms whose past innovation aligns well with the newly chosen technological standard, but that advantage is temporary.
- Over time, standards stimulate follower innovation, knowledge spillovers, technological convergence, and sectoral growth.
- Methodologically, the paper also establishes that patent text and standard text can be linked at scale in a way that yields a useful economic measure, not just a descriptive text score.

7.2 Economic intuition

Standards do two things at once. First, they choose a technology and therefore temporarily shift competition toward the firms already closest to that technology. This is the *leader advantage.* Second, they make the chosen technology legible to the rest of the industry and provide a common framework for compatibility and diffusion. This is the *spillover and catch-up channel.* The paper's central insight is that both forces are present, but on different horizons: leader rewards appear immediately, while follower catch-up dominates later.

7.3 Takeaway

The cleanest way to read this paper is that standards are neither purely monopoly-granting devices nor purely neutral coordination tools. They are both. In the short run, they tilt the market toward firms whose knowledge already fits the selected technology. In the long run, however, they release information, shape the direction of future invention, and help laggards close the gap. That is why the paper finds temporary leader gains but persistent follower-driven growth. For projects on innovation, diffusion, standard setting, industrial policy, or the competition–innovation nexus, this paper is an excellent bridge between IO and macro.